

WHITECOURT, AB, Canada

WMO: 719300

Lat: 54.140N

Lon: 115.784W

Elev: 785

StdP: 92.25

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.4	-27.0	-33.7	0.2	-30.1	-30.2	0.3	-26.6	10.8	-2.0	9.1	-3.0	2.2	300	0.648

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	27.0	17.0	25.2	16.1	23.6	15.3	18.5	24.7	17.3	23.1	16.2	21.7	3.0	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	12.6	21.3	15.0	11.7	19.8	14.0	11.0	18.6	55.8	24.6	51.7	23.4	48.3	21.7	22.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.8	7.6	6.4	-34.0	29.8	4.6	1.8	-37.3	31.2	-40.0	32.2	-42.6	33.2	-45.9	34.6		
			-34.1	20.4	4.4	1.4	-37.2	21.4	-39.8	22.1	-42.3	22.9	-45.5	23.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-10.7	-8.2	-3.5	3.8	9.9	13.8	16.3	14.9	10.2	3.7	-4.9	-10.4
	DBStd	11.35	9.09	7.78	7.34	4.92	4.32	3.07	2.98	3.23	4.19	4.52	6.68	7.82
	HDD10.0	3150	643	509	417	191	56	5	1	3	47	199	446	633
	HDD18.3	5631	901	743	676	437	263	140	77	112	244	453	696	891
	CDD10.0	584	0	0	0	4	52	118	196	156	54	4	0	0
	CDD18.3	24	0	0	0	0	1	3	14	7	1	0	0	0
	CDH23.3	402	0	0	0	1	28	56	187	103	27	1	0	0
	CDH26.7	59	0	0	0	0	3	6	33	14	3	0	0	0
Wind	WSAvg	2.6	2.6	2.6	2.7	2.9	3.0	2.7	2.6	2.3	2.4	2.5	2.4	2.5
Precipitation	PrecAvg	594	32	27	24	30	58	99	110	90	40	23	29	30
	PrecMax	770	71	56	42	74	209	224	212	157	93	64	56	65
	PrecMin	362	8	3	2	5	17	31	22	17	9	3	5	7
	PrecStd	101	18	16	13	20	43	56	56	39	23	17	14	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.6	9.8	13.5	21.6	26.7	27.5	29.8	28.6	26.6	20.4	10.9	9.0
		MCWB	4.0	3.9	5.9	10.4	13.4	16.2	18.6	17.9	16.3	11.8	5.8	3.7
	2%	DB	5.5	6.5	10.4	17.6	23.2	24.9	27.2	26.0	23.3	15.6	7.3	4.7
		MCWB	1.8	1.8	4.0	8.3	12.4	14.9	17.7	17.4	14.6	8.9	3.2	1.0
	5%	DB	3.1	4.1	8.1	14.6	20.9	22.9	25.4	24.1	20.6	12.9	4.8	2.0
		MCWB	0.5	0.6	2.6	6.7	11.3	14.1	16.9	16.4	13.5	7.0	1.8	-0.4
	10%	DB	0.8	1.8	5.9	12.1	18.7	20.9	23.6	22.2	17.8	10.5	2.7	-0.2
		MCWB	-0.8	-0.8	1.4	5.2	10.1	13.3	15.9	15.6	11.9	6.0	0.6	-1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.3	6.2	11.1	14.8	17.6	20.5	19.8	17.2	12.3	6.1	4.0
		MCDB	8.5	8.9	12.5	20.7	23.5	24.0	26.5	25.8	24.7	19.4	10.2	8.5
	2%	WB	2.2	2.1	4.4	8.7	13.2	16.1	19.0	18.3	15.3	9.5	3.8	1.4
		MCDB	5.1	6.0	9.7	16.5	20.9	22.5	25.0	24.3	21.8	14.7	6.5	4.1
	5%	WB	0.7	0.7	3.0	7.2	12.1	15.0	17.7	17.2	13.9	7.8	2.2	-0.3
		MCDB	2.8	4.0	7.7	13.9	19.4	20.9	23.8	22.5	19.5	11.9	4.3	1.9
	10%	WB	-0.8	-0.7	1.7	5.7	10.8	14.0	16.5	16.1	12.5	6.3	0.7	-1.8
		MCDB	1.0	1.8	5.5	11.1	17.6	19.2	22.1	21.1	17.3	9.8	2.5	-0.2
Mean Daily Temperature Range	5% DB	MDBR	8.7	10.1	10.3	11.0	12.0	11.2	11.3	11.6	11.4	9.4	7.4	7.9
		MCDBR	10.5	12.6	13.2	16.2	16.2	14.9	14.8	15.1	16.6	13.8	9.7	9.6
		MCWBR	7.8	8.8	8.0	8.8	7.8	6.9	6.8	7.5	8.8	8.0	6.5	7.3
	5% WB	MCDBR	10.3	12.3	12.7	15.2	14.8	12.8	13.4	14.0	15.5	12.9	8.9	9.5
		MCWBR	7.9	8.9	7.9	8.6	7.4	6.5	6.7	7.4	8.6	7.9	6.4	7.6
Clear-Sky Solar Irradiance	taub	0.247	0.256	0.285	0.327	0.349	0.358	0.358	0.358	0.317	0.284	0.261	0.236	
	taud	2.430	2.501	2.479	2.413	2.395	2.383	2.430	2.389	2.517	2.587	2.481	2.385	
	Ebn at Noon	757	875	911	904	898	890	883	860	862	827	730	691	
	Edh at Noon	49	64	83	102	110	112	105	103	80	60	47	41	
All-Sky Solar Radiation	RadAvg	0.87	1.86	3.27	4.55	5.45	5.76	5.82	4.87	3.38	1.91	0.92	0.63	
	RadStd	0.07	0.14	0.25	0.39	0.43	0.44	0.39	0.36	0.35	0.20	0.08	0.05	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page